

| <b>Section</b>         | <b>Component</b>              | <b>Signal</b> | <b>ESP32-S3<br/>GPIO /<br/>Notes</b> |
|------------------------|-------------------------------|---------------|--------------------------------------|
| Wake-Up<br>Sensors     | MQ2 Gas<br>Sensor             | D0            | GPIO 1                               |
| Wake-Up<br>Sensors     | External<br>Comparator<br>OUT | D0            | GPIO 2                               |
| I2S<br>Microphone      | INMP441                       | SCK /<br>BCLK | GPIO 4                               |
| I2S<br>Microphone      | INMP441                       | WS / LRC      | GPIO 5                               |
| I2S<br>Microphone      | INMP441                       | SD / DIN      | GPIO 6                               |
| I2S<br>Microphone      | INMP441                       | L/R           | GND (Left<br>channel<br>selected)    |
| I2C Bus<br>(RTC)       | DS3231                        | SDA           | GPIO 8                               |
| I2C Bus<br>(RTC)       | DS3231                        | SCL           | GPIO 9                               |
| I2C Bus<br>(RTC)       | DS3231                        | VCC           | 3.3V (Do<br>NOT use<br>5V)           |
| SPI Bus A<br>(SD Card) | SD Module                     | CS            | GPIO 10                              |
| SPI Bus A<br>(SD Card) | SD Module                     | MOSI          | GPIO 11                              |
| SPI Bus A<br>(SD Card) | SD Module                     | SCK           | GPIO 12                              |
| SPI Bus A<br>(SD Card) | SD Module                     | MISO          | GPIO 13                              |

|                               |                |                     |         |
|-------------------------------|----------------|---------------------|---------|
| SPI Bus B<br>(LoRa<br>SX1278) | LoRa<br>Module | MOSI                | GPIO 14 |
| SPI Bus B<br>(LoRa<br>SX1278) | LoRa<br>Module | MISO                | GPIO 15 |
| SPI Bus B<br>(LoRa<br>SX1278) | LoRa<br>Module | SCK                 | GPIO 16 |
| SPI Bus B<br>(LoRa<br>SX1278) | LoRa<br>Module | CS (NSS)            | GPIO 17 |
| SPI Bus B<br>(LoRa<br>SX1278) | LoRa<br>Module | RST                 | GPIO 18 |
| SPI Bus B<br>(LoRa<br>SX1278) | LoRa<br>Module | DIO0<br>(Interrupt) | GPIO 7  |